

Abstract No.

Rational Design of SIRT2 Inhibitors as Potential Glioblastoma Therapeutics Using Combined 3D-QSAR, Docking, and Machine Learning Methods

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Sirtuin 2 (SIRT2) is a key regulator of apoptosis, DNA repair, and the cell cycle, making it a promising target for anticancer therapy. In glioblastoma, SIRT2 has been implicated in tumor progression and treatment resistance, highlighting its potential as a therapeutic target in this malignancy. Achieving high selectivity remains challenging due to the conserved NAD⁺ binding site among sirtuin family members. Derivatives of 5-((3-amidobenzyl)oxy)nicotinamide are among the most potent and selective SIRT2 inhibitors, providing a foundation for further exploration. In this study, a 3D-QSAR model was constructed using 86 nicotinamide-based SIRT2 inhibitors, complemented by GRIND-derived pharmacophore modelling. The model, developed using partial least squares regression, achieved $R^2 = 0.89$ and $Q^2 = 0.69$, with no outliers in the applicability domain. Model interpretation highlighted the positive influence of steric features, secondary amide groups, and nitrogen atoms in the pyridine ring on SIRT2 activity. Fine-tuning the propynyl and central phenyl groups produced derivatives with predicted enhanced potency. Molecular docking on SIRT1–3 isoforms enabled the development of two machine learning classification models (Naive Bayes and k-nearest neighbours) to predict SIRT1/2 and SIRT2/3 selectivity profiles. Several selective SIRT2 inhibitors were identified *in silico*, and these will be synthesised and tested *in vitro*, supporting the development of novel SIRT2-targeted anticancer therapies.

By integrating 3D-QSAR, selectivity modelling, and ADMET predictions, several promising selective SIRT2 inhibitors were identified. These candidates will be synthesised and tested *in vitro* to validate their efficacy and selectivity, advancing the development of novel SIRT2-targeted anticancer therapies.



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